

Autonomous Retail Research Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: AAI-01

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Academic Year: 2025 -2026

Problem Statement & Objectives -

1. Problem Statement :-

The retail industry is experiencing rapid transformation driven by digital commerce, evolving consumer expectations, supply chain innovation, and global competition. Businesses require continuous and accurate market intelligence to remain competitive and make informed strategic decisions.

However, traditional retail research processes are:

- Time-consuming and manual
- Fragmented across multiple sources
- Lacking structured summarization
- Unable to maintain conversational continuity
- Not scalable for dynamic market environments

There is a strong need for an intelligent, autonomous research system that can process user queries, gather reliable web data, synthesize insights, and generate structured executive-level reports while maintaining historical knowledge for future reference.

2. Project Objectives :-

The primary objectives of this project are:

- To develop an LLM-powered Autonomous Retail Research Agent.
- To enable conversational retail market intelligence generation.
- To integrate LangChain and CrewAI for autonomous task orchestration.
- To allow the agent to navigate the web using integrated search tools.
- To synthesize information from multiple high-quality sources.
- To generate structured, professional research reports automatically.
- To persist research outputs in a knowledge repository for future retrieval.
- To provide strategic insights, risks, and opportunity analysis for retail markets.

3. Scope of the Project :-

The scope of the project includes:

- Retail market research and comparative analysis
- Growth trend identification
- Risk and opportunity assessment
- Strategic recommendation generation
- Conversational multi-turn research
- Knowledge storage and retrieval

The project focuses specifically on the retail domain and does not include:

- Real-time financial trading systems
- Consumer transaction processing
- Direct integration with enterprise ERP systems

Proposed Solution -

The proposed solution is an AI-powered Autonomous Retail Researcher Agent built using:

- Large Language Models (LLM)
- LangChain framework
- CrewAI multi-agent orchestration
- Web search integration tools
- Backend API services
- Database storage for research persistence

Working Mechanism :

- The user submits a retail-related query.
- The system retrieves prior conversation context (if available).
- The agent autonomously navigates the web using integrated tools.
- Multiple sources are analyzed and synthesized.

- The LLM generates a structured research report including:
 - Executive Summary
 - Market Trends
 - Opportunities
 - Risks
 - Key Insights
 - Strategic Recommendations
- The final report is formatted and stored in a knowledge repository.
- The user can access historical research from the dashboard.

This creates a fully automated retail intelligence pipeline.

1. Key features :-

- Autonomous Web Research Capability
- Multi-Agent Orchestration using CrewAI
- Context-Aware Conversational Memory
- Structured Executive-Level Report Generation
- Retail Market Trend Analysis
- Risk & Opportunity Identification
- Strategic Recommendation Engine
- Knowledge Repository Storage
- Search History Retrieval
- KPI-based Dashboard Analytics

2. Overall Architecture / Workflow :-

1). **User Interface Layer (Frontend)**

- Built using Streamlit
- Accepts natural language queries
- Displays structured research reports
- Shows previous research history
- Displays KPI metrics

2). Application Layer (Backend API)

- Developed using FastAPI
- Handles user requests
- Manages chat history
- Coordinates research workflow
- Stores data in database

3). Intelligence Layer (LLM + LangChain + CrewAI)

LangChain:

- Manages prompt templates
- Connects LLM with external tools
- Maintains conversation memory

CrewAI:

Coordinates multiple agents such as:

- Research Agent
- Analysis Agent
- Report Formatting Agent

LLM:

- Synthesizes data
- Generates structured insights
- Produces executive reports

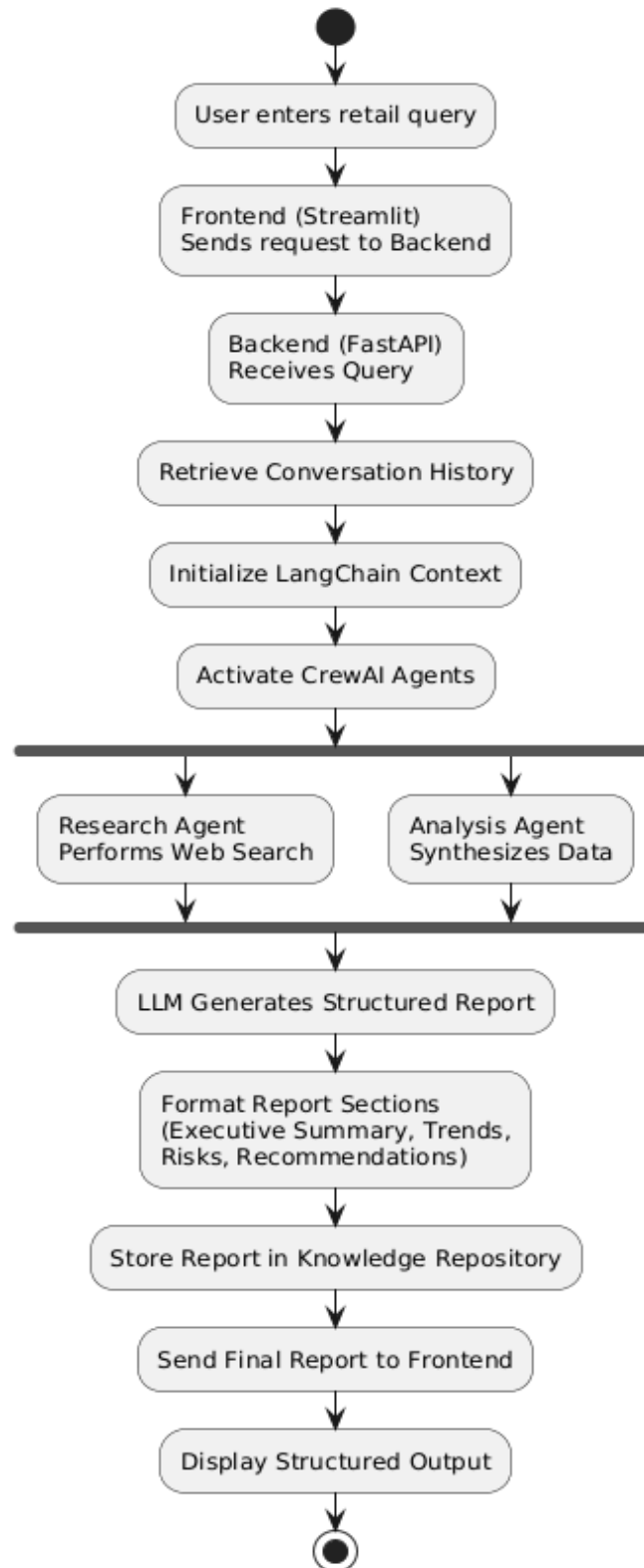
4). Data Layer (Knowledge Repository)

Stores:

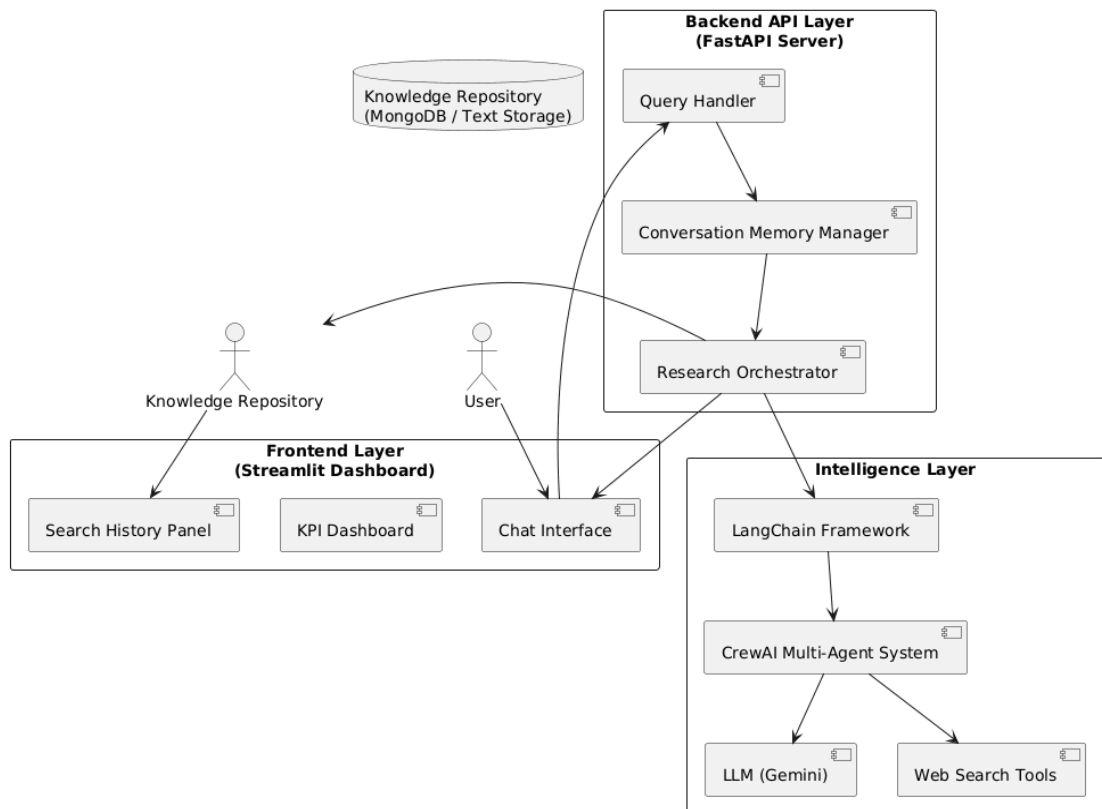
- Queries
- Reports
- Timestamps
- Conversation history

Enables retrieval of past research

Autonomous Retail Researcher Agent - Workflow



Autonomous Retail Researcher Agent - System Architecture



3. Tools & Technologies Used :-

Programming Language:

- Python

AI & Frameworks:

- Google Gemini LLM (or other LLM)
- LangChain
- CrewAI

Backend:

- FastAPI
- Uvicorn

Frontend:

- Streamlit
- HTML/CSS Styling

Database:

- MongoDB (or text-based knowledge repository)

Libraries:

- Requests
- Pandas , DateTime

Results & Output :-

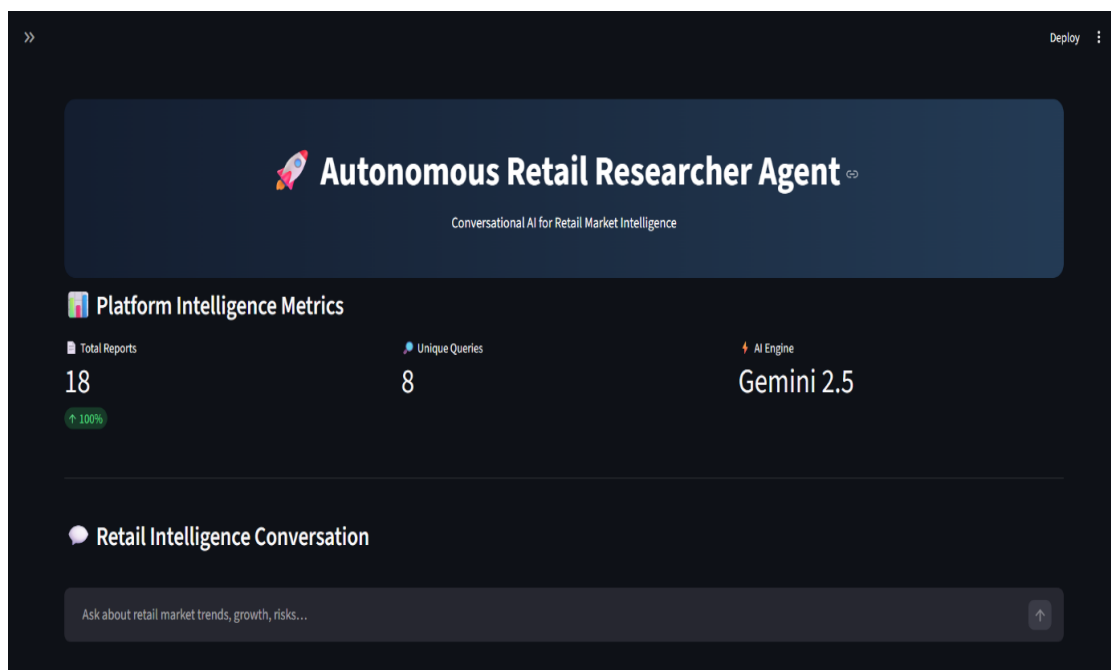
The developed system successfully:

- Processes natural language retail queries
- Conducts autonomous web research
- Synthesizes insights from multiple sources
- Generates structured professional reports
- Maintains conversational memory
- Stores research for future reference
- Displays KPI-based intelligence metrics

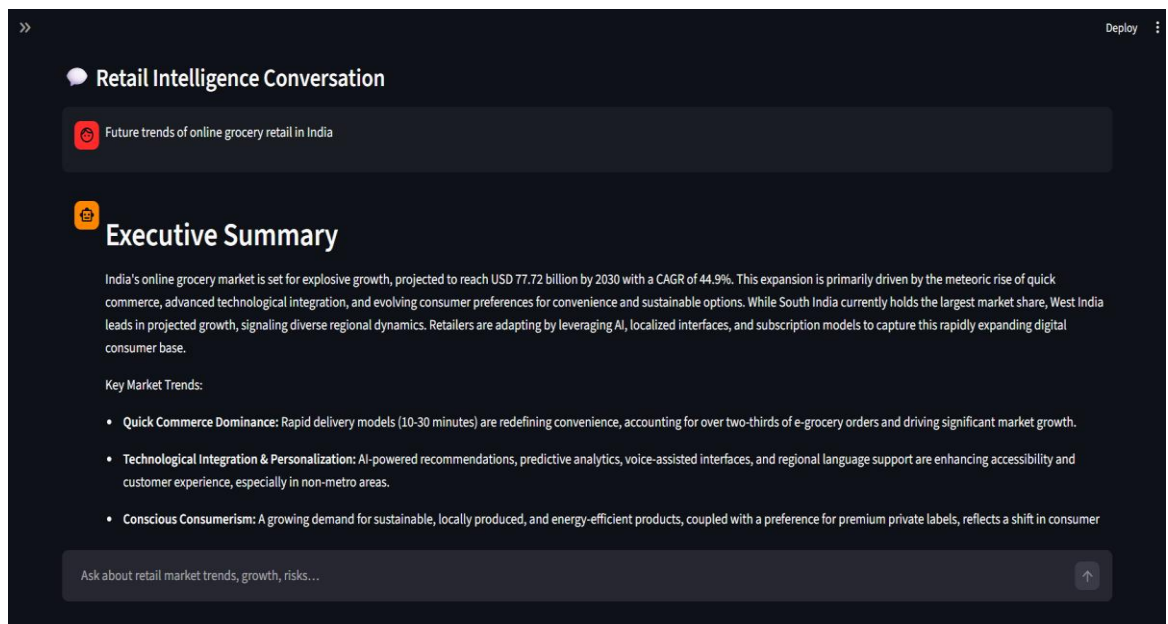
The agent demonstrates significant efficiency improvements compared to manual research workflows.

Screenshots :-

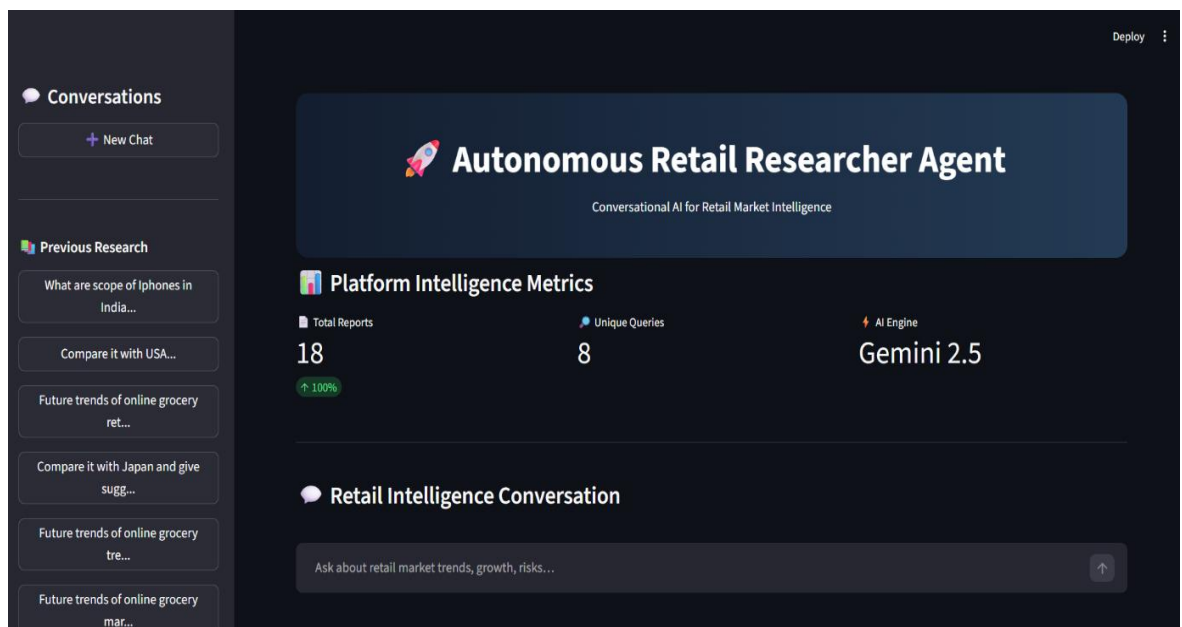
- Streamlit Dashboard Interface –



- Structured Retail Report Output –



- Search History Sidebar –



- Backend API Running –

Autonomous Retail Researcher Agent 0.1.0 OAS 3.1

/openapi.json

default

POST /research Research

GET /search-history Search History

GET /report/{report_id} Get Report

Schemas

ChatMessage > Expand all object

HTTPValidationError > Expand all object

ResearchRequest > Expand all object

ValidationError > Expand all object

- Database Collection View -

ORGANIZATION Kratin's Org - 2026-0... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin local retail_db reports sample_mflix

reports

Cluster0 > retail_db > reports

Documents 18 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query

EXPLAIN RESET FIND OPTIONS

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 18 of 18

report: "Executive Summary:
India's online grocery market is a high-growth, eme..."

chat_history: Array (3)

sources: Array (8)

embedding: Array (384)

timestamp: 2026-02-19T17:04:12.161+00:00

_id: ObjectId('699744564d15e2318953a84d')

query: "Future trends of online grocery retail in India"

report: "Executive Summary:
India's online grocery market is set for explosive ..."

chat_history: Array (1)

sources: Array (8)

embedding: Array (384)

timestamp: 2026-02-19T17:11:50.130+00:00

System Status: All Good

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Key outcomes :

- Automated retail market intelligence generation
- Reduced manual research effort by approximately 70–80%
- Structured executive-level reporting
- Improved decision support capability
- Demonstration of multi-agent AI architecture
- Practical implementation of LangChain + CrewAI

Conclusion :-

The Autonomous Retail Researcher Agent successfully integrates Large Language Models, LangChain, and CrewAI to build an intelligent, scalable, and automated research system tailored for the retail industry.

The project demonstrates how AI can transform traditional market research by enabling autonomous web navigation, structured insight synthesis, and conversational intelligence. It provides a robust decision-support tool for analyzing market trends, risks, and opportunities.

Key learnings from this project include:

- Multi-agent AI orchestration
- Prompt engineering and LLM optimization
- Backend API development
- Conversational memory implementation
- Research data persistence and retrieval

The system lays a strong foundation for AI-powered industry research automation.

Future Scope & Enhancements:-

- Structured JSON-based output instead of raw text
- Integration with real-time retail data APIs
- Advanced data visualization dashboards
- Automated SWOT analysis generation
- Sentiment analysis of retail news
- Predictive analytics for demand forecasting
- Cloud deployment (AWS / GCP / Azure)
- Role-based access control
- Expansion to other industries beyond retail